

Handover Specs

May 26

We aim to define the requirements for the actuation components.
Assume a safety factor of 2. Assume a maximum 250g load

We begin on the end effector, which needs to be capable of sustaining the load under the relatively small lever arm of the end effector from the wrist center. We also assume that the load is confined to a cube with 5cm side length. Regardless of these, the most critical joint of the last 3 is the pitch joint, which actually requires the actuator to move the load against the force of gravity. For the sake of simplicity and consistency, we assign the same spec for the second roll joint of the spherical wrist with the pitch and the first roll (elbow roll) the same specs as the elbow shoulder and base. Assuming a zero mass end effector, the wrist pitch motor needs to sustain a torque of 250g $\cdot$ 75mm = 0.01875 Nm. Assuming a 7:1 cycloidal gearbox is attached, and an efficiency of 75%, the targeted in. torque of the wrist ~~last~~ and effector motors is approximately 0.003616 Nm or 3.616 mNm. Including a safety factor and potential reductions in the cycloidal ratio due to bearing size limitations, I selected the FIT1039, which can be purchased with an encoder. The motors, gearbox, and encoders add approximately 150g to the end effector each, for a total weight at the end effector of ~700g. Including a weight of at most 200g in the link between elbow and wrist gives us a targeted elbow torque of:

$$\underbrace{(750g + 200g)}_{\text{weight}} \cdot \underbrace{15cm}_{\substack{\text{not technically} \\ \text{the COM, but} \\ \text{erring on side of caution}}} \cdot \underbrace{\frac{1}{0.75}}_{\text{efficiency}} \cdot \underbrace{2}_{\text{safety factor}} \cdot \underbrace{\frac{1}{7}}_{\text{gearbox}} = 3257 \text{ mNm}$$

The Nema 17, which has a torque of 0.25-0.35 Nm should be sufficient. The Nema 23 is selected for base and shoulder joints